Table 2.1: Comparison of Printing Techniques. The table is adapted from [1].

Parameter	Gravure	Offset	Flexographic	Screen	Inkjet	Microcontact & Nanoimprint	Transfer
Print Resolution [µm]	50-200	20-50	30-80	30-100	15-100	1-20	4-50
Print Thickness [µm]	0.02-12	0.6-2	0.17-8	3-30	0.01-0.5	0.18-0.7	0.23-2.5
Print Speed [m/min]	8-100	0.6-15	5-180	0.6-100	0.01-5	0.006-0.6	N/A
Solution Viscosity [Pa. S]	0.01-1.1	5-2	0.01-0.5	0.5-5	0.001-0.1	0.1	N/A
Material Wastage	Yes	Yes	Yes	Yes	No	Yes	No
Technique	Contact	Contact	Contact	Contact	Contactless	Contact	Contact
Printing area	Large	Large	Large	Medium	Large	Medium	Medium

of minutes. All printing techniques have material wastage during the printing process except inkjet and transfer printing methodologies. Solution viscosity is the viscosity of the ink and pastes that are compatible with the printing techniques. The technique and printing area refers to the material deposition method (contact/contactless) and substrate requirement for the applications, respectively.

Even though all printed technologies provide a variety of advantages still inkjet-printed electronics is of special interest due to its unique features such as mask-less, low-cost and non-contact deposition methods. The focus of this thesis is on inkjet-printed electronics technology, but the developed tools and concepts can also be applied to other additive technologies for PE. The detailed introduction of the inkjet-printed electronics technology is given in the next section.

2.2 Inkjet-Printed Electronics

Inkjet printing technology has gained a lot of interest and attention due to its unique feature of depositing functional inks through digital printing. The inkjet printable functional inks have enabled manufacturing of RFID tags, displays, thin-film transistors and variety of sensors [87]. Based on their electrical properties, these inks can be divided into three categories: dielectrical, semiconductive and conductive inks.

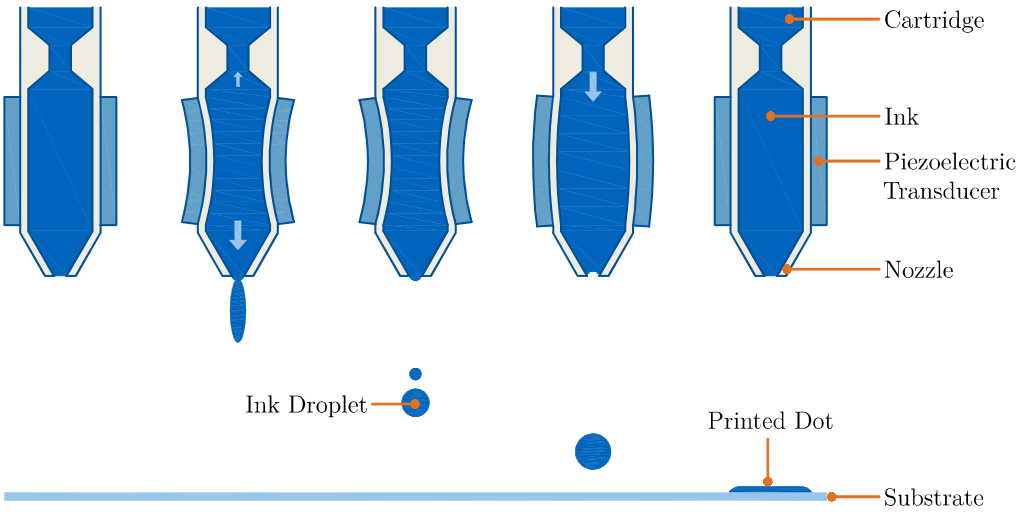


Figure 2.3: Drop-on-demand print head during printing process [6]